An Efficient Synthesis of Allenyl Perfluoroalkyl Ketones from Mono-1,2-Addition–Elimination Reaction of Allenoates with R_fMgX

Guangke He, Can Xue, Chunling Fu, Shengming Ma*

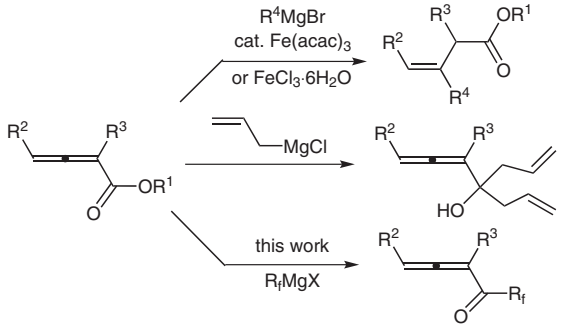
Laboratory of Molecular Recognition and Synthesis, Department of Chemistry, Zhejiang University, Hangzhou 310027, Zhejiang, P. R. of China
Fax +86(21)62609305; E-mail: masm@mail.sioc.ac.cn

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Abstract: An efficient method for the synthesis of perfluoroalkyl allenyl ketones via the 1,2-addition–elimination reaction of perfluoroalkyl Grignard reagents with allenoates has been established. No further reaction was observed between these perfluoroalkyl ketones and R_fMgX. Under the catalysis of $PdCl_2$ or $AuCl_3$, perfluoroalkyl 1,2-allenyl ketones can be transformed to 2-perfluoroalkyl-substituted furans; they may also undergo 1,2-addition with organolithiums to afford tertiary 1,2-allenols containing a perfluoroalkyl group.

Key words: 1,2-addition, elimination, allenoates, allenyl ketones, Grignard reagents, cyclization

It is well known that the reaction of carboxylates with Grignard reagents usually affords tertiary alcohols via the intermediacy of ketones and it is very difficult to stop at the stage of ketones.[1] Recently we have reported the Fe(III)-catalyzed conjugate addition reaction of 2,3-allenoates with Grignard reagents affording 3-(*Z*)-alkenoates highly regio- and stereoselectively.[2] However, when allylic Grignard reagents were used tertiary 2,3-allenols were formed.[3] Thus, we showed interest in the reaction of 2,3-allenoates with R_fMgX in the presence of Fe(III) catalyst. In this paper, we wish to disclose our recent observation that the reaction of allenoates with perfluoroalkyl Grignard reagents afforded allenyl perfluoroalkyl ketones, which may be used for the synthesis of a variety of different synthetically attractive products (Scheme 1).



Scheme 1

n-C_4F_9MgX was prepared by the I–Mg exchange reaction of n-C_4F_9I with PhMgBr in anhydrous Et_2O at –80 °C.[4] The reaction of methyl 2-methyl-4-phenyl-2,3-butadienoate (**1a**) with this prepared Grignard reagent at –70 °C in the presence of $Fe(acac)_3$ (5% mol)[2] afforded 4-phenyl-2,3-butadien-2-yl perfluorobutyl ketone (**2a**) in 48% NMR yield (entry 1, Table 1). The reaction in the absence of $Fe(acac)_3$ afforded **2a** in a much higher yield (entry 2, Table 1). It should be noted that the tertiary alcohol **3a**, 1,4-conjugate addition product **4a**, and the 1,4-conjugate addition-1,2-addition products **5a** and **6a** were not formed as judged by the analysis of the NMR spectrum of the crude product.

After literature searching, we noticed that Haszeldine et al.[5] reported the reaction of R_fMgX ($R_f = n$-C_3F_7) with RCO_2Et (R = Me, Et, *n*-Pr, CF_3, C_2F_5) afforded the mixtures of secondary alcohols, ketones, and tertiary alcohols; Ando et al.[6] recently reported the synthesis of two α-aminoalkyl perfluoroalkyl ketones from the reaction of *N*-Boc-L-phenylalanine methyl ester with R_fMgX ($R_f = n$-C_4F_9). In addition, he also reported that the reaction of (4*R*,5*R*)-2,2-dimethyl-1,3-dioxolane-4,5-dicarboxylate with R_fMgX ($R_f = n$-C_4F_9 or n-C_8F_{17}) affords the products with one carboxylate being converted to bisperfluoroalkyl-substituted tertiary alcohol and the other being converted to perfluoroalkyl ketone.[7] Furthermore, it is interesting to note that no reaction was observed in THF (entry 3, Table 1); The reaction with 3.0 equivalents of the Grignard reagents led to an incomplete conversion (entries 3–6, Table 1); even with n-$C_4F_9Ti(OPr\text{-}i)_4MgBr$, the reaction afforded ketone **2a** in 47% yield (entry 7, Table 1);[8] the yield for the reaction with ethyl 2-methyl-4-phenyl-2,3-butadienoate (**1b**) is relatively higher than that with the methyl ester (entry 8, Table 1).

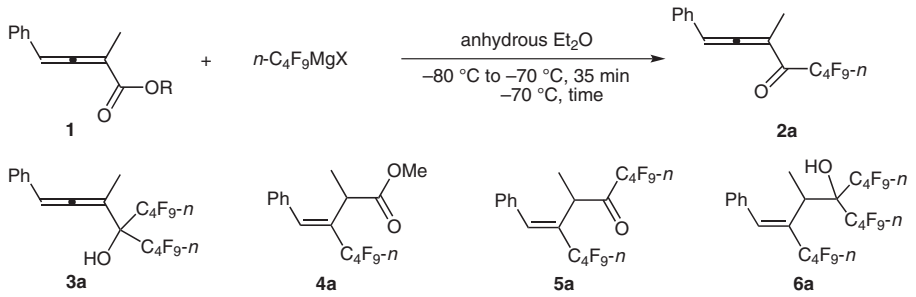
As we know that perfluoroalkyl ketones[9–12] are important in organic synthesis, pharmaceutical industry, and material science, and both starting materials are easily available,[13,14] we started to explore the scope of this reaction (Table 2). Differently substituted 2,3-allenoates, such as 2-monosubstituted (entries 8 and 9, Table 2), 2,4- (entries 1–7, Table 2), or 4,4-disubstituted (entry 10, Table 2), and full-substituted allenoates (entries 11–15, Table 2) may all undergo this transformation to afford the 1,2-allenyl perfluoroalkyl ketones in moderate yields.

The reaction may also be extended to 4,5-dodecadienoate (**1k**, Equation 1).

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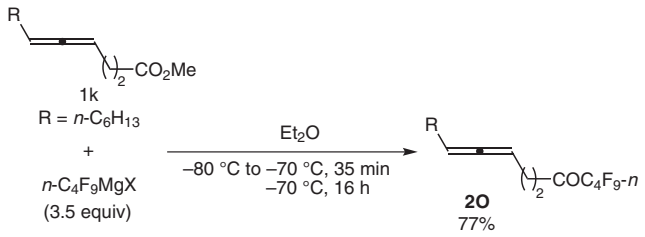
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Table 1 Optimization of Reaction Conditions for the Reaction of 2-Methyl-4-phenyl-2,3-butadienoates with *n*-C_4F_9MgX



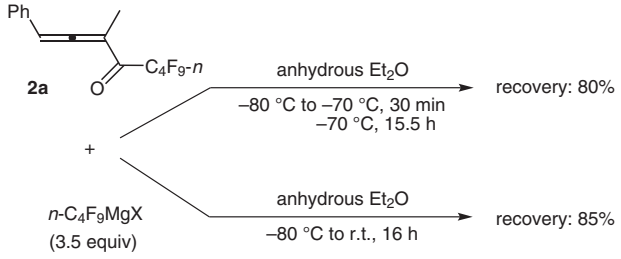
Entry	**1** R	*n*-C_4F_9MgX (equiv)	Time (h)	NMR yield of **2a** (%)	Recovery of **1a** (%)
1[a]	**1a** Me	3.0	5	48	30
2	**1a** Me	3.5	6	78	0
3[b]	**1a** Me	3.5	20.5	0	63
4	**1a** Me	3.0	16	75	2
5	**1a** Me	2.5	16	78	8
6	**1a** Me	2.0	16	58	10
7	**1a** Me	3.0[c]	11	47	0
8	**1b** Et	3.5	6	88	0

[a] The reaction was conducted at –78 °C using $Fe(acac)_3$ (10 mol%) as the catalyst.
[b] The reaction was conducted in anhyd THF.
[c] *n*-$C_4F_9Ti(Oi\text{-}Pr)_4MgX$ (3.0 equiv) was used and the reaction was conducted at –78 °C to r.t.



Equation 1

The resulting 1,2-allenyl perfluoroalkyl ketone **2a** was observed to be stable even up to room temperature towards *n*-C_4F_9MgX, explaining the results observed (Scheme 2).



Scheme 2

These 1,2-allenyl perfluoroalkyl ketones have been demonstrated to be useful intermediates in organic synthesis as shown in Scheme 3. The reaction of **2a** with MeMgCl afforded 1-perfluoroalkyl-2,3-allenols **7a** in a low NMR yield (41%); however, the reaction of **2a** with MeLi or *n*-BuLi afforded the 1,2-addition products **7a** and **7b** in 80% and 70% yields, respectively. 2-Perfluoroalkyl-substituted furans **8a** and **8b** may be formed in the presence (with 5 mol% $PdCl_2$)[16,17] and absence (with 5 mol% $AuCl_3$)[18,19] of allyl bromide (5 equiv), respectively.

In conclusion, we have developed an efficient synthesis of allenyl perfluoroalkyl ketones from the reaction of allenoates with perfluoroalkyl Grignard reagents, which is very different from the normal reaction of carboxylates with classic Grignard reagents. Due to the easy availability of the starting materials and the potential of the products, this reaction will be very useful for the introduction of perfluoroalkyl group into a variety of different organic skeletons. Further studies in this area are being pursued in our laboratory.

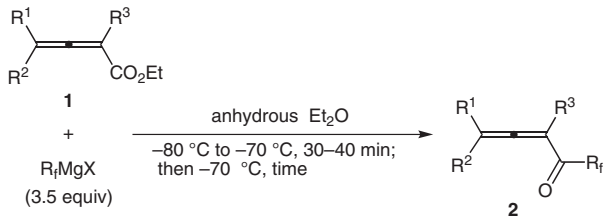
Acknowledgment

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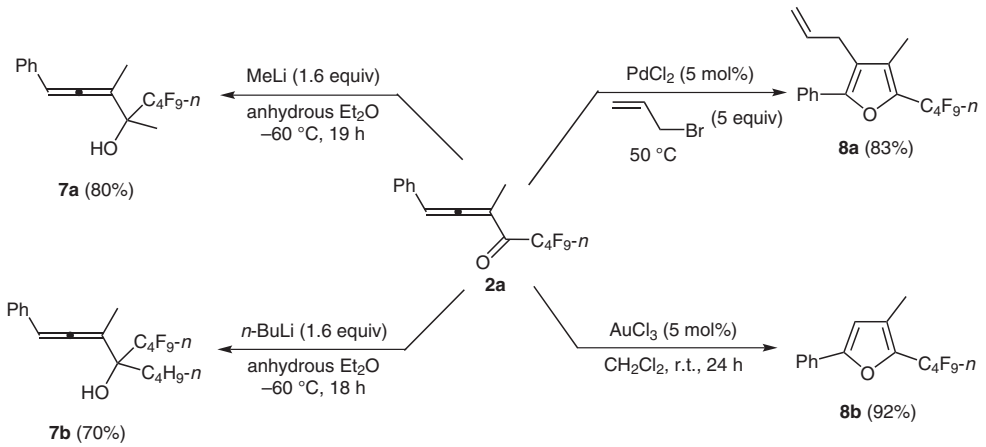
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Table 2 Reaction of 2,3-Allenoates and Perfluoroalkyl Grignard Reagents



Entry	1	R^1	R^2	R^3	R_fMgX[a] R_f =	Time (h)	Isolated yield of **2** (%)
1	**1b**	Ph	H	Me	*n*-C_4F_9 (A)	6	**2a** 74
2[b]	**1b**	Ph	H	Me	*n*-C_4F_9 (A)	23	**2a** 83
3	**1b**	Ph	H	Me	*n*-C_6F_{13} (B)	16	**2b** 70
4	**1c**	*n*-Bu	H	Me	*n*-C_4F_9 (A)	15	**2c** 66
5[c]	**1c**	*n*-Bu	H	Me	*n*-C_6F_{13} (A)	16	**2d** 63
6	**1d**	*n*-C_7H_{15}	H	Me	*n*-C_4F_9 (A)	6	**2e** 75
7[c]	**1d**	*n*-C_7H_{15}	H	Me	*n*-C_6F_{13} (A)	16	**2f** 68
8	**1e**	H	H	Bn	*n*-C_4F_9 (A)	16	**2g** 58
9	**1f**	H	H	*n*-Pr	*n*-C_6F_{13} (A)	12	**2h** 61
10[d]	**1g**	Ph	Et	H	*n*-C_4F_9 (A)	21	**2i** 53
11[e]	**1h**	$(CH_2)_5$		Me	*n*-C_4F_9 (A)	17	**2j** 74
12[c]	**1h**	$(CH_2)_5$		Me	*n*-C_6F_{13} (A)	16.5	**2k** 55
13	**1i**	Ph	Et	Me	*n*-C_4F_9 (A)	16	**2l** 71
14[c]	**1i**	Ph	Et	Me	*n*-C_6F_{13} (A)	16.5	**2m** 85
15	**1j**	Ph	Ph	*n*-Pr	*n*-C_4F_9 (B)	25	**2n** 83

[a] A: R_fMgX was prepared via EtMgBr with R_fI (1.5 equiv) in Et_2O at –80 °C; B: R_fMgX was prepared via PhMgBr with R_fI (1.5 equiv) in Et_2O at –80 °C. For detailed procedures, see footnote 15 in the reference part.
[b] The reaction was conducted with a scale of 5.0 mmol for **1b**.
[c] *n*-$C_6F_{13}MgX$ (4.0 equiv) was used.
[d] *n*-C_4F_9MgX (4.0 equiv) was used and 23% of the starting material was recovered.
[e] The reaction was conducted at –60 °C.



Scheme 3

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References and Notes

(1) (a) Hoye, T. R.; Danielson, M. E.; May, A. E.; Zhao, H. *Angew. Chem. Int. Ed.* **2008**, *47*, 9743. (b) Aponick, A.; Li, C.; Palmes, J. A. *Org. Lett.* **2009**, *11*, 121. (c) Barluenga, J.; Fernández, A.; Satrústegui, A.; Diéguez, A.; Rodríguez, F.; Fañanás, F. J. *Chem. Eur. J.* **2008**, *14*, 4153. (d) Matano, Y.; Miyajima, T.; Ochi, N.; Nakao, Y.; Sakaki, S.; Imahori, H. *J. Org. Chem.* **2008**, *73*, 5139. (e) Song, Y.; Hwang, S.; Gong, P.; Kim, D.; Kim, S. *Org. Lett.* **2008**, *10*, 269.

(2) (a) Lu, Z.; Chai, G.; Ma, S. *J. Am. Chem. Soc.* **2007**, *129*, 14546. (b) Lu, Z.; Chai, G.; Zhang, X.; Ma, S. *Org. Lett.* **2008**, *10*, 3517. (c) Chai, G.; Lu, Z.; Fu, C.; Ma, S. *Adv. Synth. Catal.* **2009**, *351*, 1946.

(3) Chen, B.; Lu, Z.; Chai, G.; Fu, C.; Ma, S. *J. Org. Chem.* **2008**, *73*, 9486.

(4) Howells, R. D.; Gilman, H. *J. Fluorine Chem.* **1975**, *5*, 99.

(5) Haszeldine, R. N. *J. Chem. Soc.* **1953**, 1748.

(6) Omote, M.; Eto, Y.; Tarui, A.; Sato, K.; Ando, A. *Tetrahedron: Asymmetry* **2007**, *18*, 2768.

(7) Sokeirik, Y. S.; Hoshina, A.; Omote, M.; Sato, K.; Tarui, A.; Kumadaki, I.; Ando, A. *Chem. Asian J.* **2008**, *3*, 1850.

(8) For a recent study on the reaction of *n*-$C_4F_9Ti(Oi\text{-}Pr)_4MgBr$ with carboxylates to afford perfluoroalkyl *sec*-alcohol, see: Mikami, K.; Murase, T.; Itoh, Y. *J. Am. Chem. Soc.* **2007**, *129*, 11686.

(9) For the reaction of perfluoroalkyl lithium reagents with ester, acyl fluoride, and anhydride affording mixtures of perfluoroalkyl-containing alcohols and ketones, see: (a) Kokotos, C. G.; Baskakis, C.; Kokotos, G. *J. Org. Chem.* **2008**, *73*, 8623. (b) Gassman, P. G.; O'Reilly, N. J. *J. Org. Chem.* **1987**, *52*, 2481.

(10) For the reaction of normal carboxylates with R_fI in the presence of MeLi–LiBr affording perfluoroalkyl ketones, see: (a) Robert, J. C.; Timothy, T. C.; William, A. M. *J. Fluorine Chem.* **1998**, *88*, 71. (b) Rong, G. *J. East China Univ. Sci. Tech.* **1996**, *22*, 356. (c) Uno, H.; Shiraishi, Y.; Suzuki, H. *Bull. Chem. Soc. Jpn.* **1989**, *62*, 2636. (d) Uno, H.; Shiraishi, Y.; Simokawa, K.; Suzuki, H. *Chem. Lett.* **1987**, *6*, 1153.

(11) For the reaction of anhydride or acyl chloride with R_fMgX reagents to afford perfluoroalkyl ketones, see: Patrice, M.; Naima, N.; Auguste, C. *J. Fluorine Chem.* **1987**, *34*, 421.

(12) For the reaction of perfluoroalkylated ester or carboxylic acid with organometallic reagents (LDA, PhLi, Al/$HgCl_2$, MeMgBr, and $NaOCH_3$) to afford the perfluoroalkyl ketones, see: (a) Reeves, J. T.; Song, J. J.; Tan, Z.; Lee, H.; Yee, N. K.; Senanayake, C. H. *J. Org. Chem.* **2008**, *73*, 9476. (b) Jiang, X.; Kakuda, K.; Matsukawa, S.; Yamamichi, H.; Kojima, S.; Yamamoto, Y. *Chem. Asian J.* **2007**, *2*, 314. (c) Konno, T.; Takehana, T.; Mishima, M.; Ishihara, T. *J. Org. Chem.* **2006**, *71*, 3545. (d) Zeifman, Y. V.; Postovoi, S. A. *J. Fluorine Chem.* **2004**, *125*, 1815. (e) Croxtall, B.; Fawcett, J.; Hope, E. G.; Stuart, A. M. *J. Fluorine Chem.* **2003**, *119*, 65. (f) Kitazume, T.; Ohnogi, T.; Lin, J. T.; Yamazaki, T. *J. Fluorine Chem.* **1989**, *42*, 17.

(13) Trost, B. M.; Pinkerton, A. B.; Seidel, M. *J. Am. Chem. Soc.* **2001**, *123*, 12466.

(14) Perfluoroalkyl iodides are commercially available.

(15) **Method A**
To a dried Schlenk tube were added *n*-C_4F_9I (0.36 mL, *d* = 2.01 g/mL, 0.72 g, 2.09 mmol) and anhyd Et_2O (2.5 mL). The resulting mixture was cooled to –80 °C in a cooling bath and a solution of EtMgBr (0.47 mL, 3.0 M in Et_2O, 1.41 mmol) was added dropwise at –80 °C with stirring. After the addition, anhyd Et_2O (0.7 mL) was used to rinse the remaining EtMgBr. The resulting mixture was stirred for 70 min at this temperature. A solution of **1b** (81.0 mg, 0.40 mmol) in anhyd Et_2O (1 mL) was added dropwise at –80 °C with stirring. Then the mixture was allowed to warm up to –70 °C within 30 min and stirred at –70 °C for 6 h. The mixture was quenched with 5 mL of a sat. aq NH_4Cl solution at –70 °C. After being warmed up to r.t. naturally, the mixture was extracted with Et_2O (3 × 20 mL). The combined organic extracts were washed with brine (5 mL) and dried over anhyd Na_2SO_4. Filtration, evaporation, and chromatography on silica gel [eluent: petroleum ether (bp 60–90 °C)] afforded **2a** (111.0 mg, 74%) as a liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.42–7.23 (m, 5 H), 6.69 (q, *J* = 2.6 Hz, 1 H), 2.03 (d, *J* = 2.6 Hz, 3 H). ^{13}C NMR (75 MHz, $CDCl_3$): δ = 217.1, 183.4 (t, *J* = 23.9 Hz), 130.6, 129.0, 128.5, 127.7, 102.8, 98.8, 14.5. ^{19}F NMR (282 MHz, $CDCl_3$): δ = –80.6 to –81.2 (m, 3 F), –113.5 to –114.0 (m, 2 F), –121.7 to –122.3 (m, 2 F), –125.1 to –125.7 (m, 2 F). MS (EI, 70 ev): *m/z* (%) = 377 (3.89) [M^+ + 1], 376 (24.69) [M^+], 129 (100). IR (neat): 3068, 3035, 2933, 1936, 1700, 1601, 1498, 1462, 1354, 1306, 1240, 1137, 1065, 1048, 1004 cm^{-1}. HRMS: *m/z* calcd for $C_{15}H_9F_9O$: 376.0510; found: 376.0517.

Method B
To a dried Schlenk tube were added *n*-$C_6F_{13}I$ (0.45 mL, *d* = 2.06 g/mL, 0.93 g, 2.09 mmol) and anhyd Et_2O (2.5 mL). The resulting mixture was cooled to –80 °C in a cooling bath, and a solution of PhMgBr (0.48 mL, 3.0 M in Et_2O, 1.44 mmol) was added dropwise at –80 °C with stirring within 5 min. After the addition, anhyd Et_2O (0.7 mL) was used to rinse the remaining PhMgBr. Then, the resulting mixture was stirred for 85 min at this temperature. A solution of **1b** (80.3 mg, 0.40 mmol) in anhyd Et_2O (1 mL) was added dropwise at –80 °C with stirring. Then the mixture was allowed to warm up to –70 °C within 30 min and stirred at –70 °C for 16 h. The mixture was quenched with a sat. aq NH_4Cl solution (5 mL) at –70 °C. After being warmed up to r.t. naturally, the mixture was extracted with Et_2O (3 × 20 mL). The combined organic extracts were washed with brine (5 mL) and dried over anhyd Na_2SO_4. After filtration, evaporation of the solvent, chromatography on silica gel [eluent: petroleum ether (bp 60–90 °C)] afforded **2b** [132.1 mg, 70%, eluent: petroleum ether (bp 60–90 °C)] as a yellow liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.44–7.24 (m, 5 H), 6.69 (m, 1 H), 2.03 (d, *J* = 2.3 Hz, 3 H). ^{13}C NMR (75 MHz, $CDCl_3$): δ = 217.1, 183.4 (t, *J* = 23.9 Hz), 130.5, 129.0, 128.5, 127.7, 102.8, 98.7, 14.6. ^{19}F NMR (282 MHz, $CDCl_3$): δ = –80.8 to –81.0 (m, 3 F), –113.5 to –113.8 (m, 2 F), –121.1 to –121.7 (m, 4 F), –122.7 to –123.1 (m, 2 F), –126.1 to –126.4 (m, 2 F). MS (EI, 70 ev): *m/z* (%) = 477 (4.78) [M^+ + 1], 476 (25.60) [M^+], 129 (100). IR (neat): 3035, 2934, 1936, 1703, 1600, 1498, 1461, 1364, 1315, 1240, 1205, 1145, 1071, 1015 cm^{-1}. HRMS: *m/z* calcd for $C_{17}H_9F_{13}O$: 476.0446; found: 476.0447.

(16) Ma, S.; Li, L. *Org. Lett.* **2000**, *2*, 941.

(17) **4-Allyl-3-methyl-2-(*n*-perfluorobutyl)-5-phenylfuran (8a)**
To a reaction tube was added **2a** (76.2 mg, 0.20 mmol), allyl bromide (128.2 mg, 1.0 mmol, purity: 97%), DMA (2 mL), and $PdCl_2$ (2.2 mg, 0.01 mmol) at r.t. Then the resulting mixture was stirred at 50 °C for 20.5 h as monitored by TLC (PE). After the reaction was complete, the mixture was quenched with H_2O (5 mL). The aqueous layer was extracted with Et_2O (5 × 15 mL) and dried over anhyd Na_2SO_4. Filtration, evaporation, and chromatography on silica gel (eluent: *n*-hexane) of the crude product afforded **8a** (69.8 mg, 83%) as a colorless liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.64–7.55 (m, 2 H), 7.47–7.31 (m, 3 H), 6.10–5.94 (m, 1 H), 5.21–5.11 (m, 1 H), 5.07–4.95 (m, 1 H), 3.40–3.32 (m, 2

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H), 2.12 (t, J = 2.3 Hz, 3 H). ^{13}C NMR (75 MHz, $CDCl_3$): δ = 152.3, 134.6, 134.4 (t, J = 29.6 Hz), 130.0, 128.7, 128.4, 127.8, 126.3, 119.6, 116.1, 27.8, 8.0. ^{19}F NMR (282 MHz, $CDCl_3$): δ = –80.9 to –81.2 (m, 3 F), –110.9 to –111.2 (m, 2 F), –123.8 to –124.1 (m, 2 F), –126.3 to –126.6 (m, 2 F). MS (EI, 70 ev): m/z (%) = 417 (8.38) [M^+ + 1], 416 (43.60) [M^+], 247 (100). IR (neat): 3085, 3064, 2930, 1640, 1582, 1494, 1448, 1415, 1352, 1236, 1206, 1135, 1101, 1075, 1030 cm^{-1}. HRMS: m/z calcd for $C_{18}H_{13}F_9O$: 416.0823; found: 416.0822.

(18) Hoffmann-Röder, A.; Krause, N. *Org. Lett.* **2001**, *3*, 2537.

(19) **3-Methyl-2-*n*-perfluorobutyl-5-phenylfuran (8b)**
To a dried Schlenk tube was added **2a** (112.1 mg, 0.30 mmol), anhyd CH_2Cl_2 (1.5 mL), and $AuCl_3$ (4.5 mg, 5 mol%) under the atmosphere of N_2 at r.t. The resulting mixture was stirred for 24 h at r.t. as monitored by TLC (petroleum ether) followed by the addition of Et_2O (20 mL). Evaporation and purification by flash chromatography on silica gel [eluent: petroleum ether (bp 60–90 °C)] afforded **8b** (103.4 mg, 92%) as a colorless liquid. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.72–7.62 (m, 2 H), 7.45–7.27 (m, 3 H), 6.56 (s, 1 H), 2.21 (t, J = 2.4 Hz, 3 H). ^{13}C NMR (75 MHz, $CDCl_3$): δ = 155.8, 134.8 (t, J = 31.7 Hz), 129.4, 128.8, 128.7, 127.6, 124.4, 109.4, 10.0. ^{19}F NMR (282 MHz, $CDCl_3$): δ = –80.9 to –81.2 (m, 3 F), –110.9 to –111.1 (m, 2 F), –123.8 to
–124.1 (m, 2 F), –126.3 to –126.6 (m, 2 F). MS (EI, 70 ev): m/z (%) = 377 (7.99) [M^+ + 1], 376 (48.23) [M^+], 207 (100). IR (neat): 3039, 2939, 1626, 1551, 1488, 1451, 1401, 1351, 1236, 1206, 1135, 1087, 1056, 1005 cm^{-1}. HRMS: m/z calcd for $C_{15}H_9F_9O$: 376.0510; found: 376.0509.

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